

Olympiad Test Paper — Mathematics (Class 7)

Chapter 4: Expressions Using Letter-Numbers

Subject	Mathematics (NCERT Class 7)
Chapter	4 — Expressions Using Letter-Numbers
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct.

Remember that $4n$ means $4 \times n$ (not $4 + n$), that only LIKE terms combine ($2a + 3b$ stays as it is, never $5ab$), and that $l + l = 2l$ (not l^2). When you substitute, keep every sign exactly as given. This chapter is about building and simplifying expressions — never about solving equations. Do not write on this paper — mark answers on your answer sheet.

1. Shabnam is always 3 years older than Aftab. Using a for Aftab's age, Shabnam's age is written as:

- | | |
|-------------|-------------|
| (A) $3a$ | (B) $a + 3$ |
| (C) $3 - a$ | (D) $a/3$ |

2. The expression $4n$ means:

- | | |
|------------------|----------------|
| (A) $4 + n$ | (B) $4 - n$ |
| (C) $4 \times n$ | (D) $n \div 4$ |

3. The value of $5m + 3$ when $m = 2$ is:

- | | |
|--------|--------|
| (A) 13 | (B) 10 |
| (C) 25 | (D) 16 |

4. Writing $7 \times k$ without the multiplication sign gives:

- | | |
|-------------|-----------|
| (A) $7 + k$ | (B) $k7$ |
| (C) $7k$ | (D) $k/7$ |

5. If $a = -4$, then the value of $10 - a$ is:

- | | |
|---------|--------|
| (A) 6 | (B) 40 |
| (C) -14 | (D) 14 |

6. The perimeter of a square of side q is given by the formula:

- (A) $q + 4$ (B) q^4
(C) $q/4$ (D) $4q$

7. The expression $l + b + l + b$ simplifies to:

- (A) $2lb$ (B) l^2b^2
(C) $4lb$ (D) $2l + 2b$

8. The sequence 4, 8, 12, 16, ... has n th term $4n$. Its 29th term is:

- (A) 116 (B) 33
(C) 120 (D) 429

9. Simplify: $5a + 3b - 2a + 7b - b$.

- (A) $12ab$ (B) $3a + 9$
(C) $3a + 11b$ (D) $3a + 9b$

10. A matchstick pattern uses $3n + 1$ sticks for the n th figure. The 10th figure needs:

- (A) 30 (B) 34
(C) 31 (D) 301

11. If $p = 3$ and $q = 5$, the value of $2pq - p^2$ is:

- (A) 39 (B) 24
(C) 21 (D) 51

12. A rectangle has length $(x + 5)$ and breadth x . Its perimeter, simplified, is:

- (A) $4x + 10$ (B) $2x + 5$
(C) $4x + 5$ (D) $x^2 + 5x$

13. The coefficient of the letter-number in the term $8y$ is:

- (A) y (B) 8
(C) $8y$ (D) 1

14. Parthiv makes L-shapes from matchsticks using 2 sticks per L. The number of sticks for n L-shapes is:

- (A) $n + 2$ (B) n^2
(C) $2n$ (D) $2 + n^2$

15. Which of these pairs are LIKE terms?

- (A) $3x$ and $3y$ (B) $5a$ and $5a^2$
(C) $7m$ and $2m$ (D) $4p$ and $4q$

16. The perimeter of a regular pentagon with side q is $5q$. When $q = 6$ cm, the perimeter is:

- (A) 11 cm (B) 56 cm
(C) 25 cm (D) 30 cm

17. Simplify: $6x + 9 - 4x - 2$.

(A) $9x$

(B) $11x$

(C) $2x + 11$

(D) $2x + 7$

18. The sum of three consecutive numbers can be written, using n for the middle number, as:

(A) $3n + 1$

(B) n^3

(C) $3n$

(D) $3n - 1$

19. If $t = 6$, the value of the expression $4t - 9$ is:

(A) 15

(B) 1

(C) 33

(D) 24

20. "5 more than twice a number x " is written as:

(A) $2(x + 5)$

(B) $2x + 5$

(C) $5x + 2$

(D) $2x - 5$

21. Simplify: $3a + 4a + 5a$.

(A) $12a$

(B) $12a^3$

(C) $60a$

(D) $12 + a$

22. If $a = 2$ and $b = 5$, the value of $3a + 2b$ is:

(A) 16

(B) 10

(C) 25

(D) 30

23. The term that is UNLIKE the others in $\{7x, -3x, x, 7\}$ is:

(A) $7x$

(B) $-3x$

(C) x

(D) 7

24. A taxi-style rule charges a fixed 50 plus 12 for each unit n . The expression for the total is:

(A) $50n + 12$

(B) $62n$

(C) $600n$

(D) $12n + 50$

25. The expression $9 - 2k$, when $k = -3$, equals:

(A) 3

(B) 6

(C) -15

(D) 15

26. Simplify: $10m - 3m + m$.

(A) $8m$

(B) $7m$

(C) $6m$

(D) $14m$

27. A shop sells pencils at price c and erasers at price d . Buying 5 pencils and 4 erasers costs:

(A) $9cd$

(B) $5c + 4d$

(C) $20cd$

(D) $4c + 5d$

28. The value of $x^2 + 3x$ when $x = 4$ is:

- (A) 19
- (B) 28
- (C) 14
- (D) 44

29. Which expression equals $l + l + l$?

- (A) l^3
- (B) $3l$
- (C) $l + 3$
- (D) $3 + l$

30. The next two terms after $5x$ in the pattern $x, 2x, 3x, 4x, 5x, \dots$ are:

- (A) $6x$ and $7x$
- (B) 6 and 7
- (C) x^6 and x^7
- (D) $25x$ and $36x$

31. Across three days a stall earns: Day1 $5c + 4d$, Day2 $3c + 6d$, Day3 $10c + d$. The simplified total is:

- (A) $28cd$
- (B) $18c + 11d$
- (C) $18cd + 11$
- (D) $11c + 18d$

32. If $a = 3$, $b = 4$ and $c = 0$, the value of $a + bc$ is:

- (A) 7
- (B) 0
- (C) 12
- (D) 3

33. Simplify: $4(2x)$.

- (A) $8x$
- (B) $6x$
- (C) $2x + 4$
- (D) $8 + x$

34. The expression "the cost of n books at 15 each, then 20 taken off" is:

- (A) $15(n - 20)$
- (B) $20 - 15n$
- (C) $15n + 20$
- (D) $15n - 20$

35. In the expression $6x - 7$, the number 6 is called the:

- (A) constant term
- (B) coefficient of x
- (C) like term
- (D) variable

36. Two expressions are EQUAL when they:

- (A) look the same on paper
- (B) have the same first letter
- (C) give the same value for every substitution
- (D) both contain a coefficient

37. Simplify: $2l + 3b + 4l - b$.

- (A) $6l + 2b$
- (B) $8lb$
- (C) $6l + 4b$
- (D) $9lb$

38. The perimeter of an equilateral triangle of side s is $3s$. If the perimeter is to be written for side $s = 12$ cm, it is:

- (A) 15 cm
- (B) 36 cm
- (C) 27 cm
- (D) 312 cm

39. If $x = 5$, the value of $2(x + 3)$ is:

- | | |
|--------|--------|
| (A) 13 | (B) 11 |
| (C) 16 | (D) 25 |

40. The sum of any two even numbers, written with letters as $2a$ and $2b$, is:

- | | |
|----------------|-------------|
| (A) $2ab$ | (B) $4ab$ |
| (C) $2(a + b)$ | (D) $a + b$ |

41. Simplify: $7p - p$.

- | | |
|----------|----------|
| (A) 7 | (B) 0 |
| (C) $7p$ | (D) $6p$ |

42. The value of $100 \div n$ when $n = 4$ is:

- | | |
|--------|---------|
| (A) 25 | (B) 400 |
| (C) 96 | (D) 104 |

43. "Twice the sum of a number and 5" is written as:

- | | |
|--------------|----------------|
| (A) $2x + 5$ | (B) $2(x + 5)$ |
| (C) $2x - 5$ | (D) $x + 10$ |

44. Simplify: $3x + 5 + 2x + 7$.

- | | |
|---------------|---------------|
| (A) $5x + 12$ | (B) $17x$ |
| (C) $5x + 35$ | (D) $6x + 12$ |

45. If $m = -2$, the value of $m^2 + 3$ is:

- | | |
|--------|--------|
| (A) 1 | (B) 7 |
| (C) -1 | (D) -7 |

46. The number of months in y years is given by:

- | | |
|--------------|--------------|
| (A) $y + 12$ | (B) $y/12$ |
| (C) $12y$ | (D) $y - 12$ |

47. Simplify: $9a - 4b - 3a + 4b$.

- | | |
|-----------|---------------|
| (A) $6a$ | (B) $6a + 8b$ |
| (C) $6ab$ | (D) $6a - 8b$ |

48. The term $3a^2$ and the term $5a$ are:

- | | |
|------------------|--------------------|
| (A) like terms | (B) equal terms |
| (C) unlike terms | (D) both constants |

49. If the perimeter of a square is $4s$, then for a square whose side is doubled to $2s$ the perimeter is:

- | | |
|-----------|----------|
| (A) $4s$ | (B) $6s$ |
| (C) $16s$ | (D) $8s$ |

50. The value of $(a + b)^2$ when $a = 1$ and $b = 2$ is:

- (A) 5
(C) 6
- (B) 3
(D) 9

51. A pattern uses $2n + 3$ tiles for the n th shape. The number of EXTRA tiles added going from the 4th shape to the 5th shape is:

- (A) 3
(C) 2
- (B) 5
(D) 13

52. Simplify: $5(x + 2) + 3x$. (Treat $5(x + 2)$ as $5x + 10$.)

- (A) $8x + 2$
(C) $5x + 12$
- (B) $8x + 10$
(D) $8x + 12$

53. The total length of a caterpillar made of n square segments, each of side 1 unit, placed in a row so the whole is n units long, plus a 2-unit head, is:

- (A) $2n$
(C) $3n$
- (B) $n + 2$
(D) $2n + 2$

54. If $x = 10$, which expression has the GREATEST value?

- (A) $x + 9$
(C) $x^2 \div 10$
- (B) $2x - 5$
(D) $3x - 8$

55. Using n for the middle number, the three consecutive numbers $(n - 1)$, n , $(n + 1)$ add up to:

- (A) $3n + 3$
(C) n^3
- (B) $3n$
(D) $3n - 3$

56. Simplify: $4a + 7 - a + 3 - 2a$.

- (A) $a + 10$
(C) $7a + 10$
- (B) $a + 4$
(D) $a - 10$

57. A square has side $(n + 1)$. Its perimeter, simplified, is:

- (A) $n + 4$
(C) $4n + 4$
- (B) $4n + 1$
(D) $n^2 + 1$

58. If $x = 2$, the values of $3x$ and $3 + x$ are, respectively:

- (A) 6 and 5
(C) 6 and 6
- (B) 5 and 6
(D) 5 and 5

59. A rule says: output = $4 \times (\text{input}) - 1$. When the input is 7, the output is:

- (A) 28
(C) 24
- (B) 27
(D) 11

60. The total earnings from selling 3 books at price b and 2 pens at price p , fully simplified, is:

(A) 5bp

(B) 6bp

(C) 3b + 2p

(D) 2b + 3p

End of question paper. Maximum marks: 240. Marking: +4 correct, -1 wrong, 0 unattempted.

Solutions — Mathematics (Class 7), Chapter 4: Expressions Using Letter-Numbers

Marking: +4 correct, -1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	11	C	21	A	31	B	41	D	51	C
2	C	12	A	22	A	32	D	42	A	52	B
3	A	13	B	23	D	33	A	43	B	53	B
4	C	14	C	24	D	34	D	44	A	54	D
5	D	15	C	25	D	35	B	45	B	55	B
6	D	16	D	26	A	36	C	46	C	56	A
7	D	17	D	27	B	37	A	47	A	57	C
8	A	18	C	28	B	38	B	48	C	58	A
9	D	19	A	29	B	39	C	49	D	59	B
10	C	20	B	30	A	40	C	50	D	60	C

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

- (B)** Shabnam's age is always Aftab's age + 3. With Aftab's age = a , Shabnam's age = $a + 3$. ($3a$ would mean three times Aftab's age; $3 - a$ reverses the relation.)
- (C)** By convention $4n$ is shorthand for $4 \times n$ — the multiplication sign is simply dropped. It does NOT mean $4 + n$; with $n = 5$, $4n = 20$, not 9.
- (A)** Substitute $m = 2$: $5m + 3 = 5 \times 2 + 3 = 10 + 3 = 13$. (Reading $5m$ as $5 + m$ gives 10 — the trap; $5(m+3)$ gives 25.)
- (C)** Dropping the $\times$ sign and writing the number before the letter gives $7k$. ($k7$ violates the number-first convention; $7 + k$ changes the operation.)
- (D)** The letter carries its whole sign: $10 - a = 10 - (-4) = 10 + 4 = 14$. (Ignoring the negative sign gives the trap 6; $10 \times (-4)$ would give -40.)
- (D)** Perimeter of a square = $4 \times \text{side} = 4q$. ($q + 4$ adds instead of multiplies; q^4 raises to a power.)

- 7. (D)** $1 + b + 1 + b = (1 + 1) + (b + b) = 21 + 2b$. The like terms $1 + 1$ make 21 , not 1^2 ; and 1 and b are unlike, so they stay separate (not $21b$).
- 8. (A)** The n th term is $4n$, so the 29th term is $4 \times 29 = 116$. (33 comes from $29 + 4$; "429" mistakenly writes the digits side by side.)
- 9. (D)** Collect like terms: a-terms $5a - 2a = 3a$; b-terms $3b + 7b - b = 9b$. Total $3a + 9b$. ($3a + 11b$ forgets to subtract the $-b$; $12ab$ wrongly merges unlike terms.)
- 10. (C)** Substitute $n = 10$ into $3n + 1$: $3 \times 10 + 1 = 30 + 1 = 31$. (Stopping at $3n = 30$ gives the trap 30; "301" writes the digits together.)
- 11. (C)** $2pq - p^2$ with $p = 3$, $q = 5$: $2 \times 3 \times 5 - 3^2 = 30 - 9 = 21$. (Forgetting to square, $30 - 3 = 27$ isn't offered; $2 \times 3 \times 5 + 3^2 = 39$ adds wrongly; $30 + \dots$ gives 51.)
- 12. (A)** Perimeter $= 2(\text{length} + \text{breadth}) = 2((x + 5) + x) = 2(2x + 5) = 4x + 10$. ($2x + 5$ forgets to double; $x^2 + 5x$ multiplies instead of adding.)
- 13. (B)** In $8y$ the number multiplying the letter-number is the coefficient, so it is 8. (y is the letter-number itself; 1 would be the coefficient of a bare y .)
- 14. (C)** Each L uses 2 sticks, so n L-shapes use $2 \times n = 2n$ sticks. ($n + 2$ adds rather than multiplies; n^2 squares.)
- 15. (C)** Like terms have the same letter part. $7m$ and $2m$ share the letter m , so they are like. ($3x$ & $3y$, $4p$ & $4q$ have different letters; $5a$ & $5a^2$ have different powers.)
- 16. (D)** Perimeter $= 5q$; with $q = 6$, $5 \times 6 = 30$ cm. ($5 + 6 = 11$ adds; "56" writes digits together; $25 = 5^2$.)
- 17. (D)** Collect like terms: x-terms $6x - 4x = 2x$; numbers $9 - 2 = 7$. Result $2x + 7$. (Merging everything into $9x$ or $11x$ wrongly adds unlike terms; $2x + 11$ adds the numbers wrongly.)
- 18. (C)** Three consecutive numbers about a middle n are $(n - 1)$, n , $(n + 1)$; their sum is $(n - 1) + n + (n + 1) = 3n$. ($3n + 1$ / $3n - 1$ keep a leftover that actually cancels.)
- 19. (A)** Substitute $t = 6$: $4t - 9 = 4 \times 6 - 9 = 24 - 9 = 15$. (24 forgets to subtract; reading $4t$ as $4 + t$ gives 1.)
- 20. (B)** "Twice a number" $= 2x$; "5 more than" it $= 2x + 5$. ($2(x + 5)$ doubles after adding 5; $5x + 2$ swaps the roles.)
- 21. (A)** All three terms are like (letter a), so $3a + 4a + 5a = (3 + 4 + 5)a = 12a$. ($12a^3$ wrongly multiplies the powers; $60a$ multiplies the coefficients.)

- 22. (A)** $3a + 2b$ with $a = 2$, $b = 5$: $3 \times 2 + 2 \times 5 = 6 + 10 = 16$. (Reading $3a$ as $3 + a$ etc. gives 10; "25" comes from 5^2 .)
- 23. (D)** $7x$, $-3x$ and x are all like terms (letter x); 7 has no letter, so it is the odd one out — a constant, unlike the others.
- 24. (D)** Fixed 50 plus 12 for each of n units $= 12 \times n + 50 = 12n + 50$. ($50n + 12$ swaps which quantity is multiplied; $62n$ adds the two numbers first.)
- 25. (D)** $9 - 2k$ with $k = -3$: $9 - 2 \times (-3) = 9 - (-6) = 9 + 6 = 15$. (Dropping the sign of k gives $9 - 6 = 3$; mis-signing gives -15 .)
- 26. (A)** All like terms (m): $10m - 3m + m = (10 - 3 + 1)m = 8m$. ($7m$ forgets the final $+ m$; $6m$ subtracts it.)
- 27. (B)** 5 pencils at c plus 4 erasers at d cost $5c + 4d$; c and d are unlike, so the terms stay separate (not $9cd$ or $20cd$). ($4c + 5d$ swaps the counts.)
- 28. (B)** $x^2 + 3x$ with $x = 4$: $4^2 + 3 \times 4 = 16 + 12 = 28$. (Forgetting to square, $4 + 12 = 16$ gives the trap 16; $14 = \dots$ mis-reads $3x$.)
- 29. (B)** $l + l + l$ is three l 's added $= 3l$. (l^3 would be $l \times l \times l$; $l + 3 / 3 + l$ add a number instead of counting the l 's.)
- 30. (A)** The pattern's n th term is nx , so after $5x$ come $6x$ and $7x$. (Dropping the letter gives 6 and 7; squaring the position gives $36x$ for the 6th.)
- 31. (B)** Add like terms across the three days. c -terms: $5c + 3c + 10c = 18c$; d -terms: $4d + 6d + d = 11d$. Total $18c + 11d$. ($28cd / 18cd + 11$ wrongly merge unlike terms; $11c + 18d$ swaps the totals.)
- 32. (D)** $a + bc$ with $a = 3$, $b = 4$, $c = 0$ means $a + (b \times c) = 3 + 4 \times 0 = 3 + 0 = 3$. (Adding all three, $3 + 4 + 0 = 7$, is the trap; $(a + b)c = 0$.)
- 33. (A)** $4(2x) = 4 \times 2 \times x = 8x$. ($6x$ adds the coefficients $4 + 2$; $2x + 4 / 8 + x$ treat the product as a sum.)
- 34. (D)** Cost of n books at 15 each $= 15n$; "20 taken off" $= 15n - 20$. ($15(n - 20)$ subtracts inside; $20 - 15n$ reverses the order; $15n + 20$ adds.)
- 35. (B)** In $6x - 7$ the number 6 multiplies the letter x , so it is the coefficient of x . (-7 is the constant term; x is the variable.)
- 36. (C)** Two expressions are equal when they give the same value for every substitution (e.g. $l + b + l + b$ and $2l + 2b$). Looking alike or sharing a first letter is irrelevant.

- 37. (A)** Collect like terms: l-terms $2l + 4l = 6l$; b-terms $3b - b = 2b$. Result $6l + 2b$. ($8lb / 9lb$ wrongly merge unlike terms.)
- 38. (B)** Perimeter of an equilateral triangle = $3s$; with $s = 12$, $3 \times 12 = 36$ cm. ($3 + 12 = 15$ adds; "312" writes digits together; $27 = 3^3$.)
- 39. (C)** $2(x + 3)$ with $x = 5$: $2 \times (5 + 3) = 2 \times 8 = 16$. ($2x + 3 = 13$ forgets to distribute the 2 over the 3; $2 \times 5 + \dots$ mis-steps to 11; $5^2 = 25$.)
- 40. (C)** $2a + 2b = 2(a + b)$, which is 2 times a whole number, hence even. The factored form shows divisibility by 2. ($2ab / 4ab$ wrongly multiply; $a + b$ drops the factor 2.)
- 41. (D)** $7p - p$ means $7p - 1p = (7 - 1)p = 6p$. (Treating $-p$ as -1 of nothing gives 7 or 0; $7p$ ignores the subtraction.)
- 42. (A)** $100 \div n$ with $n = 4$: $100 \div 4 = 25$. (400 multiplies; 96 subtracts; 104 adds.)
- 43. (B)** "The sum of a number and 5" = $(x + 5)$; "twice" it = $2(x + 5)$. ($2x + 5$ doubles only the number; $x + 10$ mis-distributes.)
- 44. (A)** Group like terms: x-terms $3x + 2x = 5x$; numbers $5 + 7 = 12$. Result $5x + 12$. ($17x / 6x + 12$ wrongly merge x's and numbers; $5x + 35$ multiplies the numbers.)
- 45. (B)** $m^2 + 3$ with $m = -2$: $(-2)^2 + 3 = 4 + 3 = 7$. (Squaring a negative gives +4; mistakenly writing -4 gives -1 or 1.)
- 46. (C)** 1 year = 12 months, so y years = $12 \times y = 12y$ months. ($y + 12$ adds; $y/12$ divides; $y - 12$ subtracts.)
- 47. (A)** Collect like terms: a-terms $9a - 3a = 6a$; b-terms $-4b + 4b = 0$. Result $6a$. ($6a + 8b / 6a - 8b$ mishandle the b-terms that actually cancel; $6ab$ merges unlike terms.)
- 48. (C)** $3a^2$ has the letter part a^2 while $5a$ has the letter part a ; different powers make them unlike terms, so they cannot be combined.
- 49. (D)** Perimeter = $4 \times \text{side}$. With side $2s$, perimeter = $4 \times (2s) = 8s$. ($4s$ keeps the old side; $6s$ adds $4s + 2s$; $16s$ squares the doubling.)
- 50. (D)** $(a + b)^2$ with $a = 1$, $b = 2$: $(1 + 2)^2 = 3^2 = 9$. (Computing $a^2 + b^2 = 1 + 4 = 5$ is the classic trap; $1 + 2 = 3$ forgets to square; $6 = 2 \times 3$.)
- 51. (C)** Tiles for the n th shape = $2n + 3$. 5th shape: $2 \times 5 + 3 = 13$; 4th shape: $2 \times 4 + 3 = 11$. Extra added = $13 - 11 = 2$ (the coefficient of n). (3 is the constant; $5 / 13$ misread the question.)
- 52. (B)** $5(x + 2) + 3x = (5x + 10) + 3x = (5x + 3x) + 10 = 8x + 10$. ($8x + 2$ mishandles the 10; $5x + 12$ forgets to distribute the 5; $8x + 12$ adds wrongly.)

53. (B) The n body-segments give a length of n units, and the head adds 2 units, so the total length is $n + 2$. ($2n$ doubles the body; $3n / 2n + 2$ over-count.)

54. (D) With $x = 10$: (A) $10 + 9 = 19$; (B) $2 \times 10 - 5 = 15$; (C) $10^2 \div 10 = 100 \div 10 = 10$; (D) $3 \times 10 - 8 = 30 - 8 = 22$. The greatest is (D) 22.

55. (B) $(n - 1) + n + (n + 1) = 3n$: the -1 and $+1$ cancel, leaving $3n$ — exactly 3 times the middle number. ($3n + 3 / 3n - 3$ keep a leftover that actually cancels.)

56. (A) Collect like terms: a -terms $4a - a - 2a = a$; numbers $7 + 3 = 10$. Result $a + 10$. ($a + 4$ mishandles the numbers; $7a + 10$ wrongly adds a coefficient.)

57. (C) Perimeter of a square = $4 \times \text{side} = 4(n + 1) = 4n + 4$. ($n + 4$ forgets to multiply the n ; $4n + 1$ distributes only over the n ; $n^2 + 1$ squares.)

58. (A) $3x$ with $x = 2$ is $3 \times 2 = 6$; $3 + x$ is $3 + 2 = 5$. So the values are 6 and 5 respectively. (Swapping them gives 5 and 6 — the $3x = 3 + x$ confusion.)

59. (B) Output = $4 \times \text{input} - 1$. With input 7: $4 \times 7 - 1 = 28 - 1 = 27$. (28 forgets the -1 ; $4 \times (7 - 1) = 24$ subtracts before multiplying; $4 + 7 = 11$ adds.)

60. (C) 3 books at price b cost $3b$; 2 pens at price p cost $2p$; b and p are unlike terms, so the total stays $3b + 2p$ (it cannot collapse to $5bp$ or $6bp$). ($2b + 3p$ swaps the counts.)

Solutions complete: 60 / 60.